

SHREYANSH SINGH

LLM Inference, Post-Training, and GPU Kernel Engineering

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Experience

Level AI

Jan 2022 – Present

Principal Machine Learning Engineer

India (Remote)

- Deployed Qwen3.6-35B-A3B with MTP speculative decoding for our low-latency virtual agent platform on four B200 GPUs; sustained 250 concurrent requests at <500 ms p95 time-to-first-token using eight CUDA-MPS-backed processes, FP8 KV cache, tuned serving parameters, and session-affinity routing for prefix-cache locality.
- Optimized Qwen3-ASR-1.7B serving with calibrated static W8A8 FP8, hand-written Triton kernel fusing Q/K RMSNorm, multi-axis RoPE, and paged KV-cache writes, and shape-aware CUDA Graphs for the audio encoder; improved throughput 38.3% and reduced p95 latency 32.3% versus BF16 with no measured quality loss.
- Built an agentic configuration-search system for in-house Llama-style SLMs across eight production services, benchmarking serving configurations on vLLM, SGLang, and TensorRT-LLM; identified up to 3× higher throughput for high-QPS services, while EAGLE-3 delivered a 1.4× serial-latency speedup for one low-concurrency service.
- Led development of agentic “AI Workers” for reasoning, search, and analytics across multi-step contact-center workflows, using tools, memory, context compaction, and failure recovery for executable SQL/TypeScript.
- Built the org-wide Llama 8B SFT → DPO stack on 8×H100 using `torch.compile`, FlashAttention-2, BFD packing, FSDP2, activation checkpointing, and async distributed checkpoints; trained a multi-task contact-center model adopted across teams and evaluated on internal benchmarks and held-out zero-shot tasks.

Mastercard - AI Garage

Aug 2020 – Jan 2022

Data Scientist

Gurugram, India

- Developed a graph-based representation learning method for transaction fraud detection, improving AUCPR by 6% over existing methods while retaining a practical training-time/performance trade-off.
- Co-developed MeTGAN, a memory-efficient tabular GAN that reduced memory use by approximately 80% versus the contemporary state of the art on high-cardinality categorical data without degrading generation quality; published at ICONIP 2021.

Selected Projects

Speculative-Decoding Algorithms for LLM Inference | B200 benchmarks

Apr – Aug 2026

- Implemented and benchmarked seven speculative-decoding strategies for Qwen2.5-7B, including trained EAGLE-3/MTP/PARD/Medusa/draft-model heads and n-gram/suffix baselines; EAGLE-3 delivered a 1.65× serial speedup and 1.35× higher batched throughput in vLLM over 100 prompts.
- Built a reproducible single-B200 benchmark harness for Qwen3.6-35B-A3B across SGLang and vLLM, sweeping MTP, EAGLE-3, DFlash, and DSpark under a fixed 8K-input/128-output-token workload; native MTP improved SGLang output throughput 23.1% over target-only decoding, with tail-latency regressions explicitly reported.

KV-Cache Compaction & Compression for LLM Inference

Apr 2026

- Built and benchmarked learned KV-cache compaction systems based on Cartridges and STILL; on Wikipedia history QA, a 1024-token cartridge reduced KV-cache memory 23.7× and prefill latency 37.8× versus full-context inference while matching semantic accuracy (1.00).
- Implemented post-training KV-cache eviction methods in `gpt-fast` for Llama-style decoding, including Attention Sink, L2-Norm compression, SnapKV, and H2O/Heavy-Hitter eviction.

Quantization-Aware Training for LLMs

Apr – Jun 2026

- Built a single-GPU QAT pipeline for Qwen3-4B with six FP8/INT8/INT4 variants, `torch.compile` training, compressed artifact export, and vLLM evaluation; on 500 NuminaMath examples, W8A8 INT8 QAT scored 20.0% accuracy versus 20.2% for BF16 and 18.8% for matched PTQ.

CUDA and Triton Kernels for Multi-LoRA Inference

Oct 2025

- Implemented CUDA and Triton kernels for Batched Gather Matrix-Vector Multiplication (BGMV) and Segmented Gather Matrix-Vector Multiplication (SGMV), enabling per-request LoRA routing within heterogeneous batches over a shared Llama-style backbone; validated correctness and benchmarked decode, prefill, and end-to-end generation against loop and grouped-BMM baselines.

Additional Technical Work

- **FlashAttention in PyTorch** (223+ stars): correctness-oriented FA1–FA4 implementations with manual backward paths, causal/masked modes, and a simplified FA3 FP8 forward path with per-tile quantization metadata.
- **Distributed Training from Scratch**: handwritten DP, FSDP, TP, FSDP+TP, and pipeline-parallel MLP training with explicit collectives and communication-overlap experiments.
- **Verifiable Multi-Turn Post-Training**: Qwen3-4B SFT + GRPO for a multi-turn Wordle environment with synthetic teacher traces, FSDP2/FlashAttention-2 training, structured rewards, and vLLM policy rollouts.
- **Annotated ML Papers** (288+ stars) and **technical blog** covering LLM systems, inference, GPU kernels, and research paper implementations.

Publications

- **MeTGAN: Memory Efficient Tabular GAN for High Cardinality Categorical Datasets**. Singh et al., ICONIP 2021.
- **CuRL: Coupled Representation Learning of Cards and Merchants to Detect Transaction Frauds**. Gramopadhye*, Singh* et al., ICANN 2021. (* equal contribution)
- **A Language Model Based Approach for Natural Language Generation**. Singh et al., ACL MSR Workshop 2018.

Education

IIT (BHU) Varanasi, India

Jul 2016 – May 2020

B.Tech. in Computer Science and Engineering

CGPA: 9.57/10.0

Technical Skills

Languages: Python, C/C++, CUDA C++, Triton

LLM Systems: PyTorch, Transformers, TRL, FSDP2, FlashAttention, `torch.compile`, CUDA Graphs, vLLM, SGLang, TensorRT-LLM

Core Areas: LLM post-training (SFT, DPO, GRPO), inference optimization, quantization, distributed training, CUDA/Triton kernel development, agentic systems, retrieval/ranking, evaluation

Infrastructure: Kubernetes, Linux, Git